

A Model of Consumer Behaviour based on Rationality and Propensity to Consume

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Abstract

This paper develops a unified model of consumer behaviour based on two fundamental variables: the degree of individual rationality $r \in [0, 1]$ and the propensity to consume $c \in [0, 1]$. The framework combines insights from economics, finance, psychology, philosophy, political science, welfare economics, international relations, warfare economics, statistics, computer science, and artificial intelligence. Rationality governs the extent to which individuals optimize intertemporal utility, while the propensity to consume determines the allocation of income between consumption and saving. The model generates a continuous spectrum between fully rational and completely irrational behaviour and allows aggregation to the macroeconomic level.

The paper ends with “The End”

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1 Introduction

Consumer behaviour is central to economics, finance, public policy, and social welfare. Traditional models assume fully rational agents maximizing utility. However, evidence from psychology, behavioural economics, neuroscience, and data science suggests that actual decision-making frequently deviates from strict rationality.

This paper introduces a two-dimensional behavioural framework:

$$r \in [0, 1]$$

representing the degree of rationality, and

$$c \in [0, 1]$$

representing the propensity to consume.

The pair (r, c) defines a behavioural state within a bounded unit square.

2 Definitions

Definition 1 (Rationality). *The degree of rationality is*

$$r \in [0, 1],$$

where

- $r = 0$ denotes completely irrational behaviour,
- $r = 1$ denotes perfectly rational behaviour.

Definition 2 (Propensity to Consume). *The propensity to consume is*

$$c \in [0, 1],$$

where

- $c = 0$ denotes complete saving,
- $c = 1$ denotes complete consumption.

Assume normalized income

$$Y = 1.$$

Then

$$C = c$$

and

$$S = 1 - c.$$

3 Behavioural State Space

The consumer state space is

$$\Omega = [0, 1]^2.$$

The extreme points correspond to distinct behavioural archetypes.

r	c	Interpretation
0	0	Irrational hoarder
0	1	Impulsive spender
1	0	Rational saver
1	1	Rational full consumer

Table 1: Extreme behavioural states

4 Microeconomic Utility Framework

Let utility be

$$U(C, S) = \alpha \ln(C + \varepsilon) + (1 - \alpha) \ln(S + \varepsilon),$$

where

$$0 \leq \alpha \leq 1$$

and

$$\varepsilon > 0.$$

Substituting

$$S = 1 - C,$$

the optimization problem becomes

$$\max_C [\alpha \ln(C + \varepsilon) + (1 - \alpha) \ln(1 - C + \varepsilon)].$$

4.1 Optimal Consumption

Differentiating,

$$\frac{\partial U}{\partial C} = \frac{\alpha}{C + \varepsilon} - \frac{1 - \alpha}{1 - C + \varepsilon}.$$

Setting

$$\frac{\partial U}{\partial C} = 0,$$

gives

$$\alpha(1 - C + \varepsilon) = (1 - \alpha)(C + \varepsilon).$$

Neglecting small ε ,

$$C^* = \alpha.$$

Therefore,

$$c^* = \alpha.$$

5 Behavioural Deviation Model

Let

$$\eta \sim U(0, 1)$$

represent behavioural impulses.

Actual consumption is

$$c = rc^* + (1 - r)\eta.$$

This yields a convex combination of rational optimization and behavioural noise.

5.1 Expectation

Since

$$E[\eta] = \frac{1}{2},$$

we obtain

$$E[c] = rc^* + (1 - r)\frac{1}{2}.$$

5.2 Variance

Since

$$\text{Var}(\eta) = \frac{1}{12},$$

it follows that

$$\text{Var}(c) = (1 - r)^2 \frac{1}{12}.$$

6 Theoretical Properties

Theorem 1. *Behaviour converges to the rational optimum as $r \rightarrow 1$.*

Proof. From

$$c = rc^* + (1 - r)\eta,$$

taking the limit,

$$\lim_{r \rightarrow 1} c = c^*.$$

Hence behaviour becomes perfectly rational. □

Theorem 2. *Behaviour becomes purely stochastic as $r \rightarrow 0$.*

Proof. Taking the limit,

$$\lim_{r \rightarrow 0} c = \eta.$$

Thus behaviour is entirely governed by behavioural impulses. □

7 Behavioural Efficiency

Define

$$B = 1 - |c - c^*|.$$

Then

$$0 \leq B \leq 1.$$

- $B = 1$ corresponds to perfect optimization.
- $B = 0$ corresponds to maximal deviation.

Substituting the behavioural model,

$$B = 1 - (1 - r)|\eta - c^*|.$$

Therefore,

$$E[B] = 1 - (1 - r)E|\eta - c^*|.$$

8 Psychological Foundations

The irrational component η captures:

1. Present bias.
2. Hyperbolic discounting.
3. Loss aversion.
4. Emotional spending.
5. Advertising influence.

6. Social conformity.
7. Herd behaviour.
8. Cognitive overload.

A possible decomposition is

$$\eta = \sum_{i=1}^n w_i b_i,$$

where

$$\sum_i w_i = 1$$

and b_i are behavioural biases.

9 Finance

Let wealth evolve according to

$$W_{t+1} = (1 + i)(W_t + Y_t - C_t).$$

Substituting

$$C_t = c_t Y_t,$$

gives

$$W_{t+1} = (1 + i) \left(W_t + (1 - c_t) Y_t \right).$$

Expected long-run wealth therefore depends inversely on average consumption.

10 Welfare Economics

Social welfare is

$$SW = \sum_{i=1}^N U_i.$$

Suppose policy improves average rationality from

$$\bar{r}_0$$

to

$$\bar{r}_1.$$

Then welfare improvement is

$$\Delta SW = SW(\bar{r}_1) - SW(\bar{r}_0).$$

Policies that reduce behavioural distortions increase welfare whenever individuals systematically deviate from their own optimal plans.

11 Political Economy

Governments influence consumption through

1. Taxation.
2. Subsidies.
3. Interest rates.
4. Welfare transfers.
5. Information campaigns.

Political incentives may encourage higher consumption to stimulate GDP growth despite long-run debt accumulation.

12 International Relations and Geopolitics

Aggregate consumption affects:

- Trade balances.
- Strategic dependence.
- Sovereign debt.
- Capital accumulation.
- Economic resilience.

Let

$$NX = X - M.$$

Consumption-driven economies often exhibit

$$M > X,$$

leading to

$$NX < 0.$$

This may create geopolitical dependence on foreign capital and supply chains.

13 Warfare Economics

National resilience can be represented as

$$R_n = f(S_n, K_n),$$

where

- S_n is aggregate savings,
- K_n is productive capital.

Since

$$S_n = 1 - \bar{c},$$

high consumption societies may possess lower strategic reserves during prolonged conflict.

14 Macroeconomic Aggregation

For N consumers,

$$\bar{C} = \frac{1}{N} \sum_{i=1}^N c_i.$$

Average rationality is

$$\bar{r} = \frac{1}{N} \sum_{i=1}^N r_i.$$

Using

$$c_i = r_i \alpha + (1 - r_i) \beta,$$

we obtain

$$\bar{C} = \beta + (\alpha - \beta) \bar{r}.$$

Proposition 1. *Aggregate consumption is linear in average rationality.*

Proof. Direct averaging yields

$$\bar{C} = \frac{1}{N} \sum_i [r_i \alpha + (1 - r_i) \beta].$$

Collecting terms,

$$\bar{C} = \alpha \bar{r} + \beta(1 - \bar{r}).$$

Hence

$$\bar{C} = \beta + (\alpha - \beta) \bar{r}.$$

□

15 Statistical Estimation

Observed data:

$$(r_i, c_i), \quad i = 1, \dots, N.$$

Regression model:

$$c_i = \beta_0 + \beta_1 r_i + \varepsilon_i.$$

Ordinary least squares estimates

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Goodness-of-fit is

$$R^2 = 1 - \frac{SSE}{SST}.$$

16 Artificial Intelligence and Machine Learning

Let features be

$$x = (r, \text{age}, \text{income}, \text{education}, \text{debt}).$$

A neural predictor estimates

$$\hat{c} = F(x; \theta).$$

Loss function:

$$L(\theta) = \frac{1}{N} \sum_i (c_i - \hat{c}_i)^2.$$

Gradient descent updates:

$$\theta_{t+1} = \theta_t - \lambda \nabla_{\theta} L.$$

17 Computer Science and Algorithms

17.1 Consumer Simulation Algorithm

Input:

 r
 alpha

Generate eta in [0,1]

c = r*alpha + (1-r)*eta

return c

17.2 Population Simulation

For each consumer i:

 draw r_i
 draw eta_i
 compute c_i

Compute average consumption

Compute average savings

Compute welfare

18 Philosophical Interpretation

The parameter r can be interpreted differently:

- Classical rationalism: logical optimization.
- Utilitarianism: welfare maximization.
- Aristotelian ethics: practical wisdom.
- Behavioural philosophy: bounded rationality.

- Existentialism: subjective choice.

The model therefore accommodates competing theories of human action.

19 Behavioural Phase Diagram

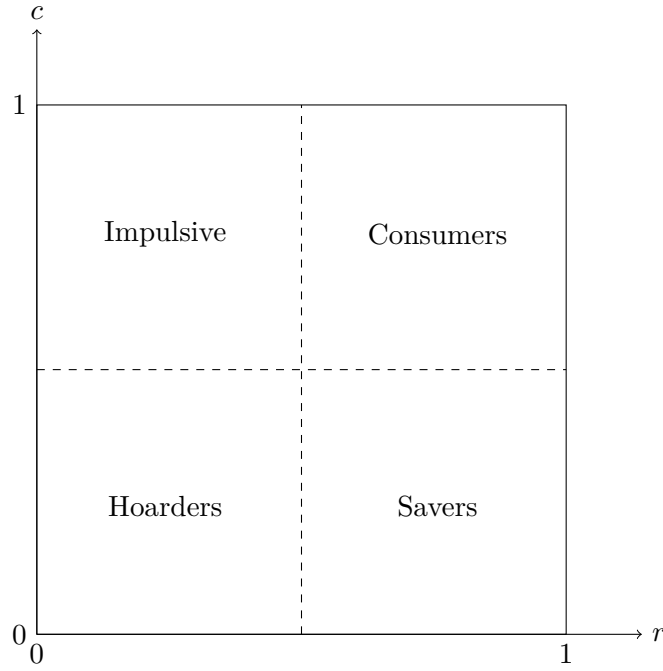


Figure 1: Consumer behavioural state space

20 Illustrative Consumption Function

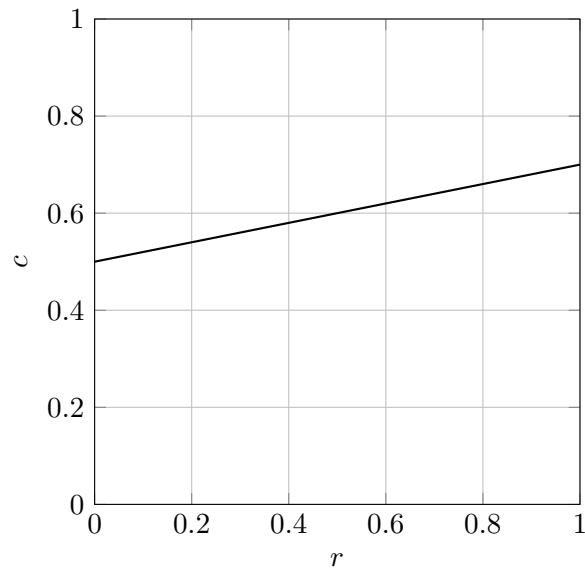


Figure 2: Expected consumption as a function of rationality

21 Conclusion

The proposed framework provides a parsimonious yet flexible representation of consumer behaviour. Rationality and consumption jointly determine individual and aggregate outcomes. The model naturally integrates economics, finance, psychology, welfare analysis, political economy, international relations, warfare economics, statistics, computer science, and artificial intelligence. Future work may incorporate network effects, endogenous rationality, adaptive learning, evolutionary dynamics, and multi-agent simulations.

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Glossary

r	Degree of rationality.
c	Propensity to consume.
C	Consumption expenditure.
S	Savings.
Y	Income.

U	Utility.
B	Behavioural efficiency.
η	Behavioural impulse variable.
α	Optimal consumption preference parameter.
SW	Social welfare.
NX	Net exports.
W_t	Wealth at time t .
R_n	National resilience.

The End